

Farès Zaghouane

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SUMMARY

My work sits between representation learning and dynamical systems. At Apple I formalized hardware behaviour as reachable-state specifications; at Nokia Bell Labs I built latent video-prediction models for cloud gaming and encountered a concrete stability problem: predictions drift once the semantic anchor becomes sparse. That is the failure mode I want to study: how much structure the predictor in a JEPA-style world model requires to remain coherent over long horizons.

RESEARCH INTERESTS

Keywords: World models · JEPA predictors · latent dynamical systems · state-space models · Koopman operators · representation learning · reinforcement learning, planning and control.

In joint-embedding predictive architectures the encoder decides what to represent while the predictor decides how representations evolve. The predictor is typically an unconstrained network. My question is whether replacing it with a structured latent dynamical system (SSM-style transitions, Koopman-style observables, or energy-based stability constraints), while holding the encoder and objective fixed, keeps long-horizon rollouts stable without expensive re-anchoring. The scope is narrow and the claim is falsifiable.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland

MSc Computer Science, Machine Learning specialization — GPA 3.7/4.0 Sept 2023 – Present

Relevant coursework: Artificial Neural Networks & Reinforcement Learning, Computer Vision, Machine Learning, Natural Language Processing, ML in Finance, Cryptography & Security.

CPE Lyon Lyon, France

Diplôme d'Ingénieur — Software Engineering & Big Data — GPA 3.6/4.0 Sept 2021 – Present

Relevant coursework: Software architecture, Design patterns, Databases, Data mining, Embedded systems, System administration, Quantum computing.

CPE Lyon Lyon, France

Licence (BSc), Computer Science & Electronics — GPA 3.6/4.0 Sept 2020 – Jul 2021

Relevant coursework: Linear algebra, Algorithms, Signal processing, Digital & analog electronics, Discrete & continuous mathematics, Quantum mechanics.

EXPERIENCE

Nokia Bell Labs Stuttgart, Germany

Research Intern (MSc Thesis) Aug 2025 – Feb 2026

- The goal was to predict dense focus maps from cloud-gaming video for bandwidth and quality allocation under strict per-frame latency constraints, using a semantic signal as teacher and a lightweight model for deployment.
- Built a teacher-student video-prediction pipeline combining Grounding DINO semantic anchoring, RAFT optical-flow propagation, and a lightweight correction network to produce dense focus maps.
- Ran experiments with JEPA-style latent prediction, state-space models, and Mamba-Vision encoders; settled on a hybrid that met the real-time latency budget while matching semantic-only accuracy closely.
- Designed the pseudo-supervision setup, tracked temporal drift across pipeline modes, and built the dense-map evaluation protocol; the hybrid recovered most of the semantic-only quality at a fraction of the cost and outperformed motion-only and saliency-only baselines.

Apple Munich, Germany

Design Verification Research Intern Oct 2022 – Jul 2023

- Modeled IP behavior using finite-state machines and event-based abstractions, allowing long-running processes to be reasoned about without expanding every low-level timing step.

- Applied model checking to behaviors that simulation cannot cover exhaustively.
- Built Python tooling that generates Verilog/SystemVerilog assertions from high-level specs and validated it on industrial designs; FSM-level specs feed directly into the model-checker.
- The event-based abstraction set the level at which reachable states are specified: model checking operates there rather than at the cycle level, which changed which properties were practical to verify.

PROJECTS

GPT-2 from Scratch

PyTorch implementation of a GPT-2-style decoder-only Transformer, built against the original papers: tokenization, token and positional embeddings, causal and multi-head attention, transformer blocks, pretraining, generation, and fine-tuning.

Stable Diffusion from Scratch

PyTorch reconstruction of latent diffusion built against the original papers: VAE encoder/decoder, CLIP-style text conditioning, U-Net denoiser, DDPM sampler, and generation pipeline.

Vision Transformer from Scratch

PyTorch Vision Transformer built from scratch around the original ViT paper: patch embeddings, learned/sine-cosine/RoPE positional encodings, self-attention blocks, and shared supervised, MAE, and DINO-style training pipelines.

Machine Learning from Scratch

From-scratch ML curriculum: 52 course-style notebooks and a 123-module scikit-learn-style library. Each chapter takes one model apart rather than only re-implementing it: why the family was invented, what every term of the equation and every hyperparameter controls, and what the experiment shows when the model breaks. The recurring question is why one design supersedes another — convolution against dense layers on images, ViT against CNN, score-based models against GANs. Runs from a custom Tensor/autograd engine and NumPy estimators through Transformers, state-space models, GNNs and generative modeling (VAE, GAN, energy-based models, DDPM, flow matching, latent diffusion) to the geometry of representations: Riemannian optimization, optimal transport, information geometry, hyperbolic embeddings, equivariance.

RL from Scratch

From-scratch PyTorch RL curriculum: 25 algorithms from tabular Q-learning to PPO/SAC, Dreamer, MuZero, and an action-conditioned JEPA world model inspired by V-JEPA 2-AC. Each method is taught through a course-style notebook, then refactored into clean package code on a shared runtime with YAML configs, run artifacts, and benchmark tooling. HalfCheetah-v5 runs reached $\approx 10k$ return for SAC/TD3 and $\approx 11k$ for DDPG around 1M steps.

V-JEPA2 ROI Semantic Extraction

Research prototype that organizes V-JEPA-style predictive video representations around semantic regions rather than uniform spatial patches.

VLM Data Engine

Video data pipeline covering shard processing, FPS conversion, scene detection, clip extraction, annotation, and quality checks; built to support multimodal world-model experiments.

SKILLS

ML & representation learning: PyTorch, TensorFlow, Neural networks, Self-supervised learning, JEPA / world models, Energy-based models

Vision & multimodal systems: Computer vision, Video understanding, Grounding DINO, RAFT / optical flow, Focus-map prediction, Multimodal data pipelines

Dynamics, control & RL: Reinforcement learning, State-space models, Mamba, Koopman operators, Dynamical systems, Physics-informed AI

Mathematical & scientific foundations: Linear algebra, Applied statistics, Signal processing, Optimization, Spectral methods, Functional analysis

Software & formal methods: Python, NumPy, Jupyter, Formal verification, Model checking, Finite-state

